JAN-P-564

19 FEBRUARY 1948

JOINT ARMY-NAVY SPECIFICATION

PIGMENT, SYNTHETIC, RED (OINTMENT TYPE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective departments and supersedes the following specification:

U. S. Army
97–54–402
18 June 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATION

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATIONS

RR–S–366—Sieves; Standard, Testing.
TT–P–141—Paint, Varnish, Lacquer, and Related Materials; General Specification for Inspection, Sampling, and Testing.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

B. TYPE

B–1. This specification covers one type of red synthetic pigment.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The pigment shall be composed of a synthetic iron oxide manufactured from a good commercial grade of iron salts.

C–2. *Workmanship.*—All work shall be done in a manner consistent with the best commercial practice in the manufacture of a high-grade pigment of the type specified.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. The material shall comply with the detail requirements listed in table I.

TABLE I.—*Detail requirements.*

	Minimum	Maximum
Total iron (as Fe_2O_3) ....(percentage)	99.0	....
Ferrous oxide .....(percentage)	....	0.50
Water-soluble matter .....(percentage)	....	0.20
Water-soluble iron as Fe (percentage)	....	0.05
Trace elements (calculated as metal)		
Arsenic[1] (ppm.)	....	10
Copper[1] (ppm.)	....	40
Lead[1] (ppm.)	....	20
pH	5.0	9.0
Moisture .....(percentage)	....	1.0
Organic colors and lakes	....	none
Coarse particles (retained by No. 325 U. S. Standard sieve[2])....(percentage)	....	1.00
Grit	....	none

[1] The manufacturer shall, at the time of forwarding his bid, submit for approval by the procuring agency, a complete description of the methods of analysis he proposes to use in the certification required in paragraph F–5f.

[2] Sieves shall conform to Federal Specification RR–S–366.

E–2. *Color.*—The mass color and tinting strength of the pigment shall be substantially the same as that of a standard sample. (See par. H–2.)

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—Unless otherwise specified in the contract or order, each manufacturing batch shall constitute a lot. (See par. H–4.)

F–3. *Sampling.*—Five containers shall be selected at random from the lot for sampling purposes. If there are fewer than five containers in the lot, all containers shall be sampled. Withdraw a ½-pound representative sample from each container so chosen. Place the sample in a clean, dry container and mix thoroughly to form a composite sample representative of the lot. Seal and label the container to insure the identity of the sample with the lot represented. (See par. H–5.)

F–4. *Material.*—The Government inspector shall obtain from the manufacturer certification that the material was manufactured from a good commercial grade of iron salts.

F–5. *Tests.*—All tests shall be conducted in accordance with the procedure outlined herein. Analytical reagent grade chemicals and

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distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents as used in the tests, and corrections shall be applied when significant.

F–5a. *Special reagents.*—

F–5a(1). *Manganous sulfate solution.*—Prepare this solution to contain 67 gm. manganous sulfate ($MnSO_4.4H_2O$), 138 ml. orthophosphoric acid (sp. gr. = 1.7) and 130 ml. sulfuric acid (sp. gr. = 1.82), diluted to 1 liter.

F–5a(2). *Stannous chloride solution.*—Prepare this solution to contain 227 gm. stannous chloride and 190 ml. hydrochloric acid (sp. gr. = 1.2), diluted to 1 liter.

F–5b. *Total iron.*—Weigh approximately 0.2000 gm. of sample into a porcelain crucible of at least 50-ml. capacity and maintain at a dull red heat for at least 1 hour. Cool, add 5 ml. concentrated hydrochloric acid, cover with a watch glass, and place on a steam bath. As the yellow color of ferric chloride forms, decolorize by adding the stannous chloride solution (see par. F–5a(2)) dropwise until all the material is dissolved and the solution remains colorless. Add, all at once, 25 ml. of a saturated solution of mercuric chloride and allow to stand for 5 minutes covered with a watch glass. Wash into a casserole containing 25 ml. of manganous sulfate solution (see par. F–5a(1)) and 350 ml. of water and titrate immediately with 0.1*N* potassium permanganate solution.

$$\text{Percentage total iron as } Fe_2O_3 = \frac{\text{ml. } KMnO_4 \times \text{normality} \times 7.985}{\text{weight of sample}}$$

F–5c. *Ferrous oxide.*—Weigh approximately 2.00 gm. of sample into a 100-ml. Erlenmeyer flask, displace the air from the flask with a continuous stream of carbon dioxide and add 10 ml. of concentrated hydrochloric acid. Digest on a steam bath until dry, then add 25 ml. of boiling water. Agitate until complete solution is obtained. Then discontinue the flow of carbon dioxide, wash into a casserole containing 25 ml. of manganous sulfate solution and 350 ml. of water and titrate immediately with 0.1*N* potassium permanganate solution.

$$\text{Percentage ferrous oxide} = \frac{\text{ml. } KMnO_4 \times \text{normality} \times 7.2}{\text{weight of sample}}$$

F–5d. *Water-soluble matter.*—Transfer 2.5 gm. of sample to a flask graduated at 250 ml. Add 100 ml. of water, boil for 5 minutes, cool, fill to the mark with water, mix and allow to settle. Pour the supernatant liquid through a dry filter paper and discard the first 20 ml. Measure 100 ml. of the filtrate into a tared dish, evaporate to dryness on the steam bath, heat for 1 hour at 105° to 110° C., cool in a desiccator and weigh. Reserve the remainder of the filtrate for the test specified in paragraph F–5e.

$$\text{Percentage water-soluble matter} = \frac{\text{weight of residue} \times 100 \times 2.5}{\text{weight of sample}}$$

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F–5e. *Water-soluble iron.*—Pipette 25 ml. of the filtrate (see par. F–5d) into a small beaker, add 1 ml. of 30-percent hydrogen peroxide, boil down to 10 ml., cool, and transfer to a comparison tube containing 2.0 ml. concentrated hydrochloric acid and 10 ml. of a 5-percent potassium sulfocyanate solution. Dilute to 50 ml. and compare the color developed by the sample with the color developed in similar preparations to which known quantities of ferric iron have been added. Estimate the water-soluble iron content in terms of percentage iron.

F–5f. *Trace elements.*—The manufacturer shall certify that the pigment meets the requirements covering maximum arsenic, copper, and lead content, when tested in accordance with methods submitted to and approved by the procuring agency.

F–5g. The pigment shall be tested for compliance with the requirements and in accordance with the test methods listed in table II.

TABLE II.—*Pigment test methods.*

Requirements	Fed. Spec. TT–P–141 test method No.
pH	511
Moisture	407
Organic colors and lakes	502
Coarse particles	410
Color:	
Mass	421
Tinting strength	422.1

F–5h. *Grit.*—Place a quantity of the sample on a smooth glass plate and pulverize with a smooth steel blade. Consider as grit any rough foreign particles which tend to give the characteristic tactile sensation experienced when sand is ground between steel and glass in the same manner.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—Unless otherwise specified in the contract or order, the material shall be packed in fiber drums, containing 50 pounds net weight, complying with the requirements of Specification 21A of Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Articles, etc.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipping containers for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Material covered by this specification is intended for use as a component of protective ointment.

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H–2. The standard sample will be furnished by the procuring agency.

H–3. Requests, requisitions, schedules, and contracts or orders should specify the title, number, and date of the specification.

H–4. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–5. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion and with the concurrence of the contractor, may modify, in whole or in part, the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H–6. Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. may be obtained from the Interstate Commerce Commission, Washington 25, D. C.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department". Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Department of the Army and the Department of the Air Force will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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